Empirical Results Report

Miratrix

2025-02-19

Introduction

This report generates the results reported in our paper (but not the figures—see the script `final_paper_figures.R` for those).

Some statistics on the studies

Our count of studies is the following:

	Blocked	Unblocked
Curated	11	0
Public	15	23

Our count of outcomes is the following:

	Blocked	Unblocked
Curated	28	0
Public	22	38

Some statistics about the blocked studies across the 50 outcomes they entail:

statistic	mean	median	sd	min	Q10	Q90	max
J	483.74	56.50	1637.23	16.00	30.00	495.00	8375.00
Jcv	0.40	0.33	0.30	0.00	0.09	0.85	1.00
K	35.16	16.00	65.39	5.00	5.00	85.30	389.00
R2.1	0.21	0.17	0.17	0.00	0.02	0.44	0.58
R2.2	0.43	0.37	0.32	0.00	0.06	0.83	0.95
R_ate	0.06	0.04	0.06	0.00	0.01	0.14	0.30
R_se	1.33	1.25	0.38	1.04	1.06	1.63	3.46
block_ICC	0.09	0.07	0.10	0.00	0.00	0.21	0.39
cluster_ICC	0.09	0.09	0.06	0.00	0.01	0.15	0.28
n	7761.36	2148.50	15702.56	409.00	854.40	17499.30	102869.00
nbar	60.41	25.37	110.50	1.74	10.19	86.35	588.04
ncv	0.41	0.34	0.21	0.01	0.21	0.60	1.35
p.tx	0.55	0.50	0.10	0.40	0.49	0.74	0.80

The 10th and 90th percentile number of individuals, clusters and blocks are around: 850 to 17,500 (individuals), 30 to 495 (clusters), and 5 to 85 (blocks).

unit	10%	90%
n	854.4	17499.3
J	30.0	495.0
K	5.0	85.3

Which studies have singletons and doubletons?

This table lists the number of singletons and doubletons for the Blocked experiments

UniqueID	num_singletons	num_doubletons
Beacon_S8TGPA	27	13
Beacon_S8TOTCR	27	13
Beacon_SRPRST2	27	13
EMERGE_team2___BB	2	0
EMERGE_eclsb2_BB	2	0
EMERGE_team2___BBSSR	2	0
EMERGE_eclsb2_BBSSR	2	0
CARES_DC_PT_Incred	36	8
CARES_TR_BPI_Incred	36	8
CARES_TR_WRS_Incred	36	8
CARES_DC_EMIT_PTH	36	8
CARES_DC_CST_COMPTOT_PTH	36	8
CARES_TR_SSRS_PTH	36	8
CARES_DC_PT_Tools_Mind	36	8
CARES_TR_WRS_Tools_Mind	36	8
CLC_GRCCSS_POST	14	1
CLC_PCCRERCORE	14	1
MathPD_MTSS	0	1
MathPD_MTSS_FRACTDEC	0	1
MathPD_MTSS_PCTRATIO	0	1
MPC_eclsb2_scale	1	25
MPC_wj_ap2	1	25
PowerTeaching_TSTSSTAND	0	2
Success_S14WJLWI_SCORE	0	4
TennStar_gktmathss	37	87
TennStar_gktreadss	37	87
TRIAD_rema2	2	3
TRIADScale-up_rema2	16	0
jg_crt_2	0	0
jg_crt_6	769	7
jg_crt_7	0	12
jg_crt_8	0	12
jg_crt_10	0	0
jg_crt_11	0	12
jg_crt_12	0	12
jg_crt_16	0	0
jg_crt_17	12	9
jg_crt_18	12	9
jg_crt_19	12	9
jg_crt_23	25	5
jg_crt_29	0	0
jg_crt_30	0	0

UniqueID	num_singletons	num_doubletons
jg_crt_31	79	13
jg_crt_34	0	0
jg_crt_41	0	0
jg_crt_52	508	0
jg_crt_56	5	5
jg_crt_63	161	9
jg_crt_68	79	13
jg_crt_69	79	13

So the number of outcomes that had no singletons is

blocked	Collection	n_outcome	n_no_singletons	n_no_doubletons
Blocked	Curated	28	5	5
Blocked	Public	22	11	8
Unblocked	Public	38	38	38

Studies with low degrees of freedom

In this report, we define "Low" to be strictly less than 10 degrees of freedom. Here are all studies with 30 or fewer "conservative" degrees of freedom defined as:

$$df_{cons} = J - 2K - 1$$

UniqueID	cons_df	n	J	K	Jbar	Jcv	ncov.1	ncov.2
TRIADScale-up_rema2	0	409	16	8	2.00	0.00	1	1
jg_crt_52	0	102869	508	254	2.00	0.00	2	2
CARES_DC_PT_Incred	8	1298	52	22	2.36	0.33	1	1
CARES_TR_BPI_Incred	8	1318	52	22	2.36	0.33	1	1
CARES_TR_WRS_Incred	8	1320	52	22	2.36	0.33	1	1
CARES_DC_EMIT_PTH	8	1229	52	22	2.36	0.33	1	1
CARES_DC_CST_COMPTOT_PTH	8	1206	52	22	2.36	0.33	1	1
CARES_TR_SSRS_PTH	8	1276	52	22	2.36	0.33	1	1
CARES_DC_PT_Tools_Mind	8	1268	52	22	2.36	0.33	1	1
CARES_TR_WRS_Tools_Mind	8	1293	52	22	2.36	0.33	1	1
jg_crt_23	9	926	41	16	2.56	0.43	2	2
jg_crt_63	9	8987	179	85	2.11	0.15	4	4
CLC_GRCCSS_POST	10	9557	28	9	3.11	0.78	2	2
CLC_PCCRERCORE	10	16465	28	9	3.11	0.78	2	2
jg_crt_7	16	1386	30	7	4.29	0.18	11	11
jg_crt_8	16	1335	30	7	4.29	0.18	11	11
jg_crt_11	16	2588	30	7	4.29	0.18	12	12
jg_crt_12	16	2612	30	7	4.29	0.18	12	12
TRIAD_rema2	24	1208	38	7	5.43	0.10	1	1
jg_crt_6	24	15993	802	389	2.06	0.42	15	15
MathPD_MTSS	27	2132	39	6	6.50	0.19	1	1
MathPD_MTSS_FRACTDEC	27	2132	39	6	6.50	0.19	1	1
MathPD_MTSS_PCTRATIO	27	2132	39	6	6.50	0.19	1	1
Success_S14WJLWI_SCORE	27	1631	37	5	7.40	0.74	1	1

Some statistics on our estimators

For each outcome we calculate a bunch of ATE estimates and SE estimates. We then look at the range of ATE estimates and the ratio of the max to min SE estimate.

We can then tally the number of outcomes with a large ranges of SE (2x or more) and/or ATE (0.10 or more) is the following (*** means yes it is a large range, blank means no):

```
##              big SE ratio
## big ATE range    ***
##                40   1
##           ***  8   1
```

How often the ATE range is large

Here we count the number of ranges greater than 0.10 and 0.05:

group	measure	n	n_10	big_10	n_05	big_5
all	ate	50	9	0.18	19	0.38
estimand	ate_Cluster	50	3	0.06	11	0.22
estimand	ate_Person	50	3	0.06	8	0.16
noFI	ate_noFI	50	8	0.16	16	0.32
noFI_estimand	ate_Cluster_noFI	50	3	0.06	9	0.18
noFI_estimand	ate_Person_noFI	50	3	0.06	3	0.06

How often estimated Standard error ratio is large

The `n_1.2`, etc are the number of outcomes with a ratio of that amount or larger. The "big" is just the percent of outcomes.

group	measure	n	n_1.2	big_1.2	n_1.5	big_1.5	n_2	big_2	max
all	se	50	29	0.58	8	0.16	2	0.04	3.46
estimand	se_Cluster	50	20	0.40	3	0.06	1	0.02	3.45
estimand	se_Person	50	17	0.34	5	0.10	0	0.00	1.75
noFI	se_noFI	50	20	0.40	5	0.10	0	0.00	1.71
noFI_estimand	se_Cluster_noFI	50	12	0.24	2	0.04	0	0.00	1.62
noFI_estimand	se_Person_noFI	50	12	0.24	3	0.06	0	0.00	1.56

Stability connected to df

FALSE CLAIM: When DF are low, the ratio of the largest to smallest standard error is always greater than 1.5x. When DF are high, only 6 out of 34 outcomes have a ratio above 1.5.

Note: The above claim is no longer true, now that we have fixed the Fully Interacted issue.

Here we divide by low vs high degree of freedom, and count how many ratios are large vs. small:

low_df	group	measure	n	n_1.2	big_1.2	n_1.5	big_1.5	n_2	big_2
low	all	se	16	16	1.00	6	0.38	1	0.06
high	all	se	34	13	0.38	2	0.06	1	0.03
low	estimand	se_Cluster	16	13	0.81	2	0.12	1	0.06
high	estimand	se_Cluster	34	7	0.21	1	0.03	0	0.00
low	estimand	se_Person	16	6	0.38	4	0.25	0	0.00

low_df	group	measure	n	n_1.2	big_1.2	n_1.5	big_1.5	n_2	big_2
high	estimand	se_Person	34	11	0.32	1	0.03	0	0.00
low	noFI	se_noFI	16	13	0.81	3	0.19	0	0.00
high	noFI	se_noFI	34	7	0.21	2	0.06	0	0.00
low	noFI_estimand	se_Cluster_noFI	16	7	0.44	1	0.06	0	0.00
high	noFI_estimand	se_Cluster_noFI	34	5	0.15	1	0.03	0	0.00
low	noFI_estimand	se_Person_noFI	16	6	0.38	3	0.19	0	0.00
high	noFI_estimand	se_Person_noFI	34	6	0.18	0	0.00	0	0.00

Fully interacted estimator notes (now defunct)

We originally had two claims:

1) To consider how much trouble the fully interacted models are causing, we exclude them and recalculate the ratios (see Figure 5). Doing so dramatically changes the story. For example, the largest estimated standard error is 1.5x the smallest estimated standard error in only 10 percent of cases without the fully-interacted models, down from 48 percent when considering all estimators. The ratio never exceeds 1.7 once the fully-interacted models are excluded.

2) Even in Figure 3 we see estimated standard errors that are 20 percent different across estimators in over 20 percent of the cases we examined.

They are no longer relevant or true.

When are SE estimators defined?

While this model-free approach is appealing in theory, in practice this means design-based standard errors can only be estimated if every block X experimental arm has at least two or more clusters. Only three studies in our sample meet this condition; consequently, the design-based standard errors do not exist for XX out of 50 outcomes when the fully-interacted model (for individual or aggregate data) is implemented. The same limitation applies to heteroskedastic robust standard errors. For most outcomes, there are only 13 estimated standard errors, rather than 19 (See Supplemental Excel Spreadsheet 1).

This table counts the number of undefined estimators within each group given the set of covariates included.

Estimator	n	n_ATE_all	n_ATE_best	n_ATE_none	n_SE_all	n_SE_best	n_SE_none
LRa-FE-db	50	0	0	0	2	0	0
LRa-FE-het	50	0	0	0	8	0	0
LRa-FIcw-db	50	12	2	0	40	34	34
LRa-FIcw-het	50	12	2	0	41	34	34
LRapw-FE-db	50	0	0	0	2	0	0
LRapw-FE-het	50	0	0	0	4	0	0
LRapw-FIpw-db	50	12	2	0	40	34	34
LRapw-FIpw-het	50	12	2	0	40	34	34
LRi-FE-crve	50	0	0	0	0	0	0
LRi-FE-db	50	0	0	0	2	0	0
LRi-FIpw-crve	50	10	0	0	40	34	34
LRi-FIpw-db	50	12	2	0	40	34	34
LRicw-FE-crve	50	0	0	0	0	0	0
LRicw-FE-db	50	0	0	0	2	0	0
LRicw-FIcw-crve	50	10	0	0	40	34	34
LRicw-FIcw-db	50	12	2	0	40	34	34
MLM-FE	50	0	0	0	0	0	0

Estimator	n	n_ATE_all	n_ATE_best	n_ATE_none	n_SE_all	n_SE_best	n_SE_none
MLM-FIcw	50	0	0	0	0	0	0
MLM-RE	50	0	0	0	0	0	0

This table counts the number of estimates for each outcome that are undefined (the different columns correspond to adjusting for all covariates, a single best covariate, or no covariates):

UniqueID	n_ATE_all	n_ATE_best	n_ATE_none	n_SE_all	n_SE_best	n_SE_none
Beacon_S8TGPA	0	0	0	8	8	8
Beacon_S8TOTCR	0	0	0	8	8	8
Beacon_SRPRST2	0	0	0	8	8	8
CARES_DC_CST_COMPTOT_PTH	0	0	0	8	8	8
CARES_DC_EMIT_PTH	0	0	0	8	8	8
CARES_DC_PT_Incred	0	0	0	8	8	8
CARES_DC_PT_Tools_Mind	0	0	0	8	8	8
CARES_TR_BPI_Incred	0	0	0	8	8	8
CARES_TR_SSRS_PTH	0	0	0	8	8	8
CARES_TR_WRS_Incred	0	0	0	8	8	8
CARES_TR_WRS_Tools_Mind	0	0	0	8	8	8
CLC_GRCCSS_POST	8	0	0	13	8	8
CLC_PCCRERCORE	8	0	0	12	8	8
EMERGE_eclsb2_BB	0	0	0	8	8	8
EMERGE_eclsb2_BBSSR	0	0	0	8	8	8
EMERGE_team2___BB	0	0	0	8	8	8
EMERGE_team2___BBSSR	0	0	0	8	8	8
MPC_eclsb2_scale	0	0	0	8	8	8
MPC_wj_ap2	0	0	0	8	8	8
MathPD_MTSS	8	0	0	10	0	0
MathPD_MTSS_FRACTDEC	8	0	0	10	0	0
MathPD_MTSS_PCTRATIO	8	0	0	9	0	0
PowerTeaching_TSTSSTAND	8	0	0	8	0	0
Success_S14WJLWI_SCORE	0	0	0	0	0	0
TRIADScale-up_rema2	6	6	0	8	8	8
TRIAD_rema2	0	0	0	8	8	8
TennStar_gktmathss	8	0	0	9	8	8
TennStar_gktreadss	8	0	0	9	8	8
jg_crt_10	0	0	0	0	0	0
jg_crt_11	0	0	0	0	0	0
jg_crt_12	0	0	0	0	0	0
jg_crt_16	0	0	0	0	0	0
jg_crt_17	0	0	0	10	8	8
jg_crt_18	0	0	0	9	8	8
jg_crt_19	0	0	0	8	8	8
jg_crt_2	0	0	0	0	0	0
jg_crt_23	0	0	0	8	8	8
jg_crt_29	0	0	0	2	0	0
jg_crt_30	0	0	0	0	0	0
jg_crt_31	0	0	0	8	8	8
jg_crt_34	0	0	0	0	0	0
jg_crt_41	0	0	0	0	0	0
jg_crt_52	6	6	0	8	8	8
jg_crt_56	0	0	0	8	8	8

UniqueID	n_ATE_all	n_ATE_best	n_ATE_none	n_SE_all	n_SE_best	n_SE_none
jg_crt_6	0	0	0	8	8	8
jg_crt_63	0	0	0	8	8	8
jg_crt_68	0	0	0	8	8	8
jg_crt_69	0	0	0	8	8	8
jg_crt_7	8	0	0	8	0	0
jg_crt_8	8	0	0	8	0	0

The range of ATE and ratio of SE estimates by study

We can look at specific outcomes to see the range of ATE estimates and the ratio of SE estimates for that outcome. The "N" column shows number of defined estimates comprising the given range.

UniqueID	num_singletons	ATE_avg	ATE_N	SE_avg	SE_N
jg_crt_12	0	0.62	19	0.09	19
jg_crt_11	0	0.58	19	0.10	19
TRIADScale-up_rema2	16	0.58	13	0.13	11
TRIAD_rema2	2	0.51	19	0.09	11
CARES_DC_EMIT_PTH	36	0.37	19	0.06	11
CARES_TR_SSRS_PTH	36	0.19	19	0.12	11
TennStar_gktreadss	37	0.19	19	0.05	11
TennStar_gktmathss	37	0.19	19	0.05	11
EMERGE_eclsb2_BB	2	0.18	19	0.10	11
jg_crt_23	25	0.16	19	0.07	11
jg_crt_6	769	0.15	19	0.02	11
EMERGE_team2___BB	2	0.15	19	0.13	11
jg_crt_8	0	0.13	19	0.09	19
jg_crt_7	0	0.13	19	0.07	19
MathPD_MTSS_PCTRATIO	0	0.12	19	0.12	19
CARES_DC_CST_COMPTOT_PTH	36	0.12	19	0.07	11
jg_crt_56	5	0.11	19	0.10	11
EMERGE_team2___BBSSR	2	0.11	19	0.13	11
MathPD_MTSS	0	0.10	19	0.13	19
MPC_eclsb2_scale	1	0.10	19	0.09	11
jg_crt_16	0	0.10	19	0.07	19
CARES_TR_WRS_Incred	36	0.10	19	0.12	11
jg_crt_68	79	0.09	19	0.03	11
jg_crt_2	0	0.08	19	0.10	19
MPC_wj_ap2	1	0.08	19	0.08	11
MathPD_MTSS_FRACTDEC	0	0.08	19	0.13	19
jg_crt_10	0	0.08	19	0.05	19
CLC_GRCCSS_POST	14	0.06	19	0.05	11
Success_S14WJLWI_SCORE	0	0.06	19	0.06	19
jg_crt_17	12	0.05	19	0.03	11
jg_crt_18	12	0.03	19	0.03	11
jg_crt_29	0	0.03	19	0.01	19
jg_crt_31	79	0.03	19	0.02	11
Beacon_S8TOTCR	27	0.02	19	0.04	11
jg_crt_19	12	0.02	19	0.04	11
jg_crt_30	0	0.02	19	0.02	19
Beacon_S8TGPA	27	0.01	19	0.05	11

UniqueID	num_singletons	ATE_avg	ATE_N	SE_avg	SE_N
CARES_DC_PT_Tools_Mind	36	0.00	19	0.07	11
EMERGE_eclsb2_BBSSR	2	0.00	19	0.12	11
Beacon_SRPRST2	27	-0.01	19	0.05	11
CARES_DC_PT_Incred	36	-0.02	19	0.08	11
CARES_TR_WRS_Tools_Mind	36	-0.02	19	0.10	11
jg_crt_69	79	-0.02	19	0.03	11
CLC_PCCRERCORE	14	-0.03	19	0.04	11
CARES_TR_BPI_Incred	36	-0.03	19	0.14	11
PowerTeaching_TSTSSTAND	0	-0.04	19	0.06	19
jg_crt_63	161	-0.05	19	0.03	11
jg_crt_52	508	-0.09	13	0.03	11
jg_crt_41	0	-0.15	19	0.08	19
jg_crt_34	0	-0.20	19	0.08	19

What is the average range and ratio?

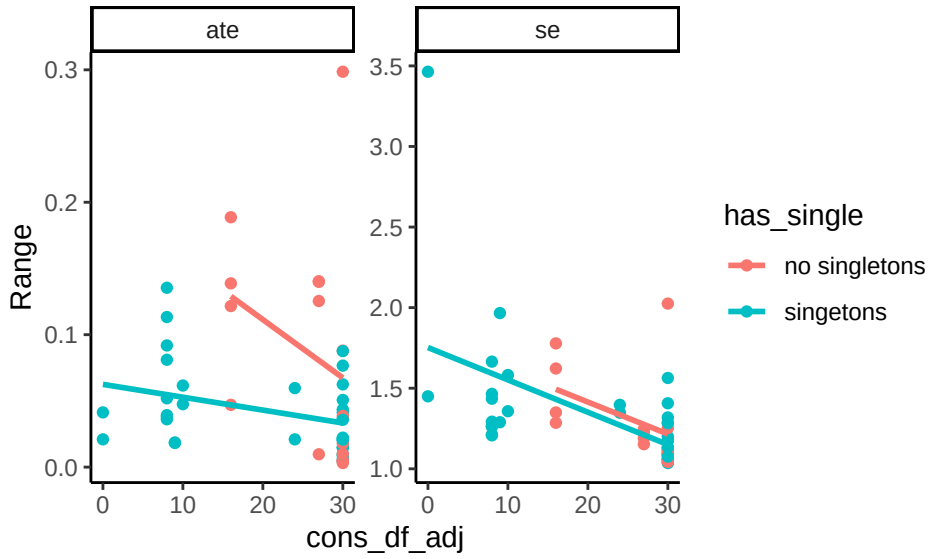
The following table shows the average range (for ATEs) and average ratio (for estimated SEs) across all our outcomes:

This tells us, for example, that "On average, the largest standard error is around 33% larger than the smallest, across studies."

group	Covariates	measure	avg	sd	Q05	Q95	n
all	best	ate	0.06	0.06	0.00	0.14	50
all	best	se	1.33	0.38	1.05	1.88	50
estimand	best	ate_Cluster	0.03	0.04	0.00	0.10	50
estimand	best	ate_Person	0.03	0.04	0.00	0.11	50
estimand	best	se_Cluster	1.25	0.36	1.04	1.56	50
estimand	best	se_Person	1.17	0.18	1.01	1.55	50
noFI	best	ate_noFI	0.05	0.06	0.00	0.14	50
noFI	best	se_noFI	1.22	0.18	1.04	1.60	50
noFI_estimand	best	ate_Cluster_noFI	0.03	0.04	0.00	0.10	50
noFI_estimand	best	ate_Person_noFI	0.02	0.03	0.00	0.09	50
noFI_estimand	best	se_Cluster_noFI	1.15	0.15	1.02	1.45	50
noFI_estimand	best	se_Person_noFI	1.14	0.15	1.00	1.47	50

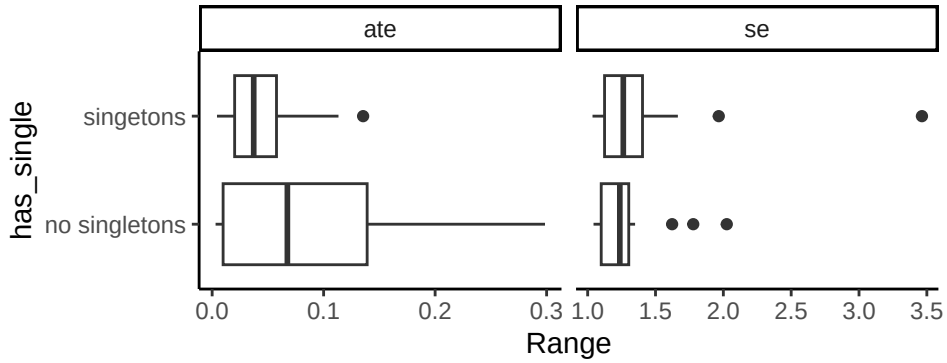
ATE range and SE ratios vs study characteristics

Here is our "conservative df" for each study and the range across estimators and the ratio of SE estimates (all studies with 30 df or more have been topcoded at 30 to make the plots easier to read):



Commentary: This plot does show there is some tendency for ranges and ratios to go down as DF goes up, but there is a lot of variation around that tendency.

Here we look at the distribution of range/ratio of estimates across outcome, dividing by singleton status. We see the studies with singletons tend to have smaller ATE ranges, and the range of SE estimates is marginally higher—note that the singleton studies SE ratios are over fewer estimators (none of the FI estimators), which would generally push the ratios *lower* as there are fewer points to take the max and min from.



Commentary: I am not sure if these plots are useful or informative.

Variation in the ATE Estimates

Studies with a high range (more than 0.15) of ATE estimates

Project	UniqueID	K	J	n	cons_df	R_ate	R_ate_Clu	R_ate_Per
Kim/set_2	jg_crt_2	7	83	2174	69	0.30	0.20	0.05
Kim/set_7	jg_crt_8	7	30	1335	16	0.19	0.11	0.18

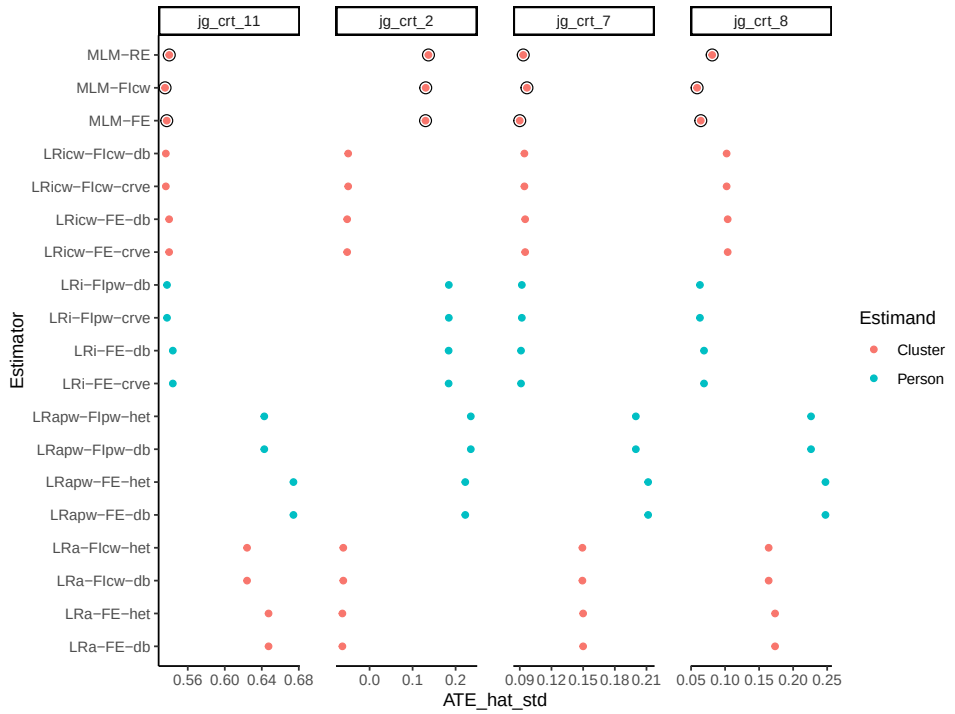
High range within Estimand

Still a threshold of 0.15.

Project	UniqueID	K	J	n	cons_df	R_ate	R_ate_Clu	R_ate_Per
Kim/set_2	jg_crt_2	7	83	2174	69	0.30	0.20	0.05
Kim/set_7	jg_crt_7	7	30	1386	16	0.12	0.06	0.12
Kim/set_7	jg_crt_8	7	30	1335	16	0.19	0.11	0.18
Kim/set_9	jg_crt_11	7	30	2588	16	0.14	0.11	0.14

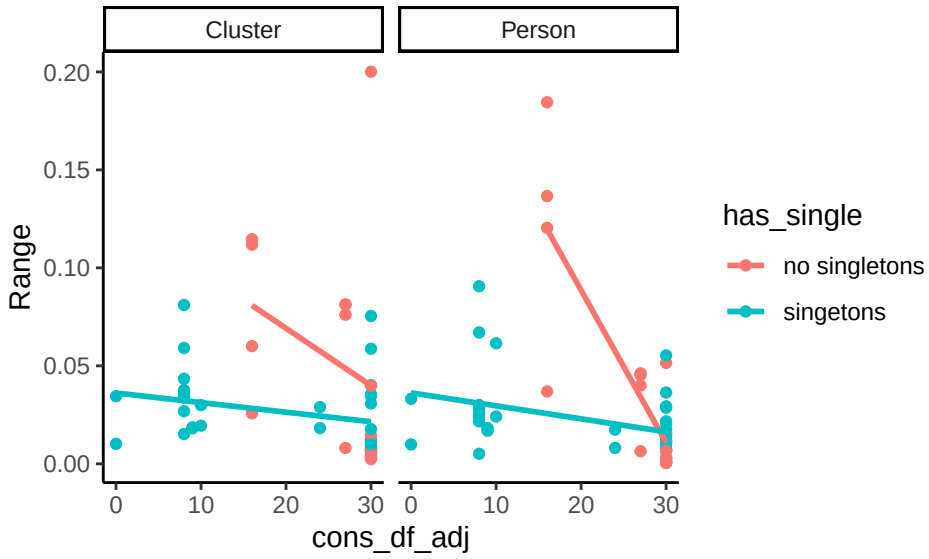
Some statistics on the selected studies:

UniqueID	num_singletons	ncv	Jcv	ATE_min	ATE_max	range	SE_avg	n_est	n_def
jg_crt_11	0	0.35	0.18	0.54	0.67	0.14	0.10	19	19
jg_crt_2	0	1.35	0.41	-0.06	0.24	0.30	0.10	19	19
jg_crt_7	0	0.51	0.18	0.09	0.21	0.12	0.07	19	19
jg_crt_8	0	0.52	0.18	0.06	0.25	0.19	0.09	19	19

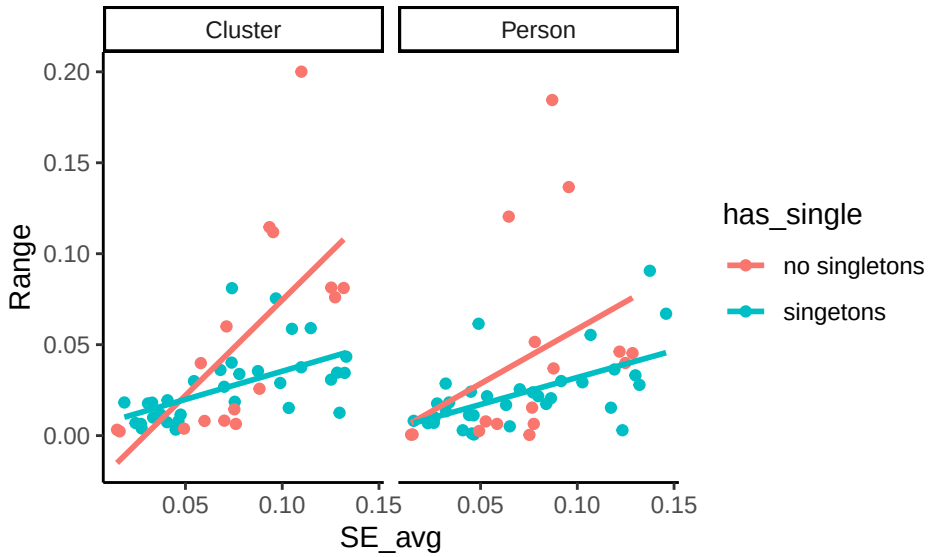


Note: The MLM estimators are circled to indicate they are not unbiased for either estimand, in principle.

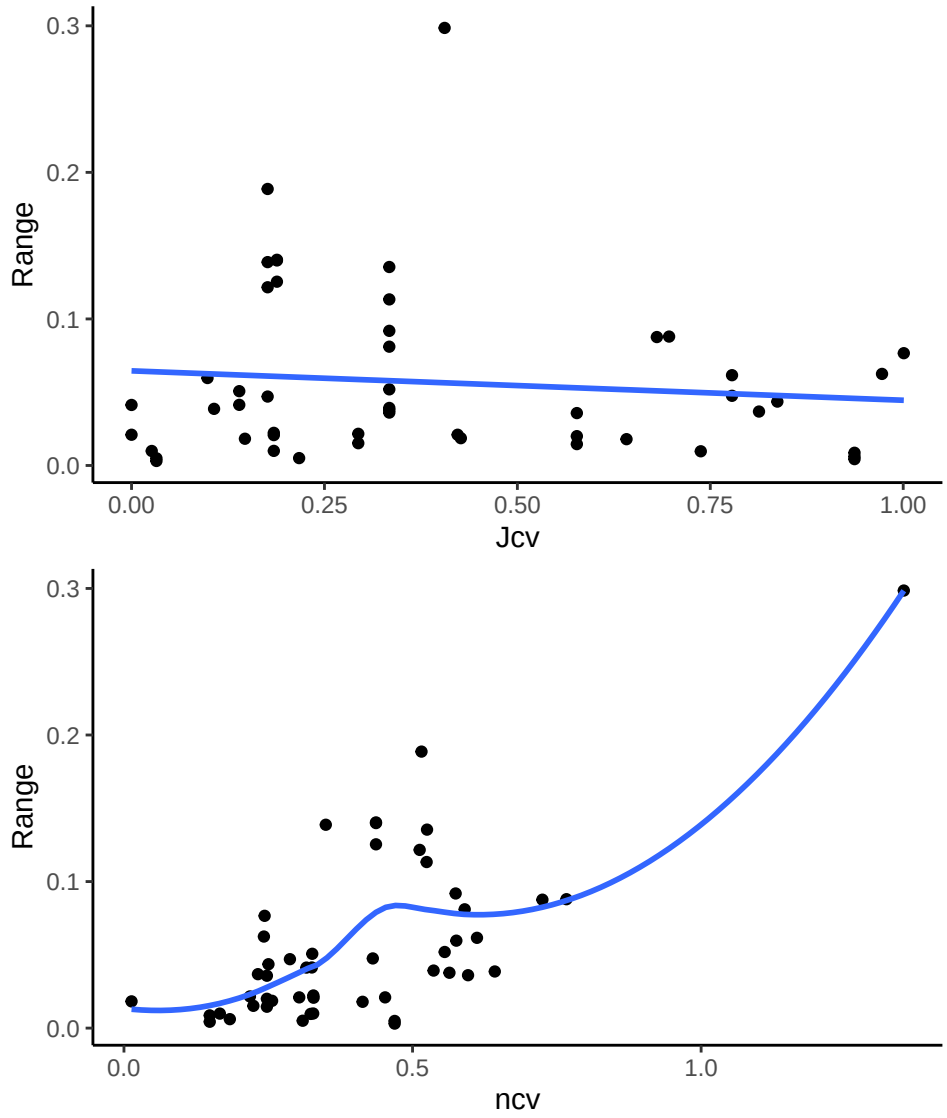
We can look at what study characteristics drive larger or smaller ranges, within estimand:



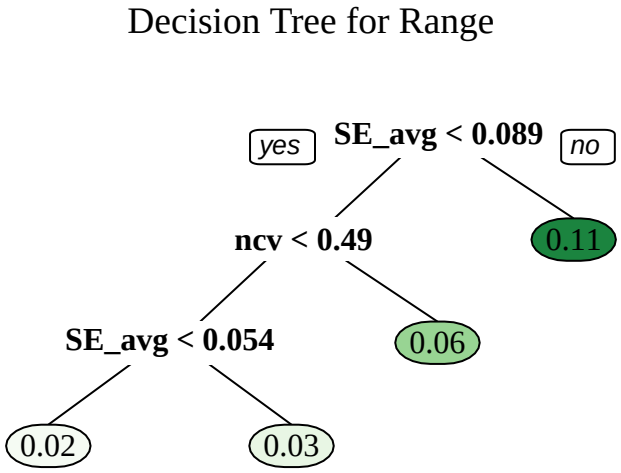
Range does go up as expected with average SE increasing:



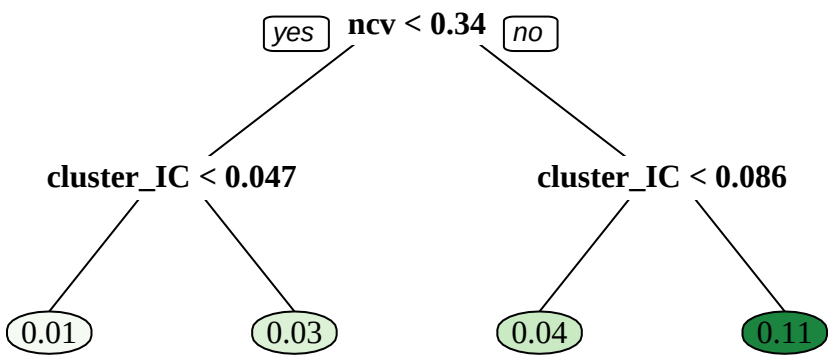
We can see if range depends on variation in block or cluster size:



We can use trees to predict what predicts larger range:

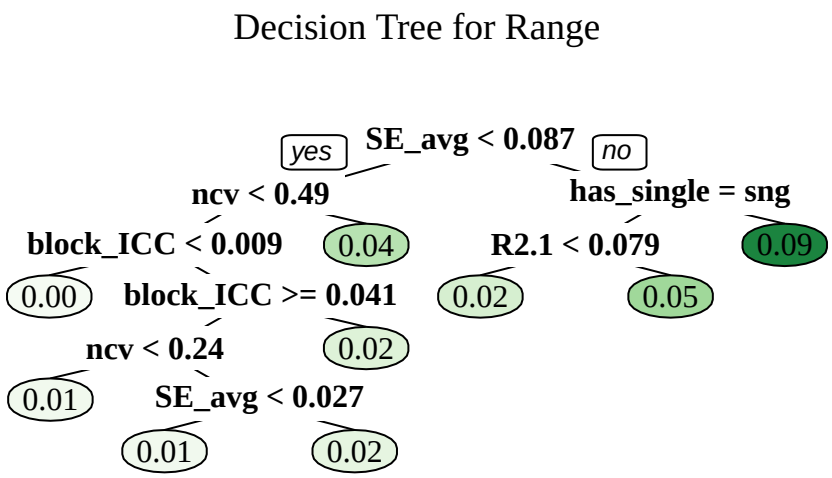


If we drop the average SE we get this:



These trees shows that uncertainty drives larger differences, and also that greater variation in cluster size drives larger differences, especially if there is variation in the cluster means.

If we look within estimand (i.e., fit our tree to the full set of 50 person ranges and 50 cluster ranges), we see that our datasets with larger R2.1 and no singletons have larger ranges (assuming the SEs are generally large):



Comparing low DF to high DF studies

We can look to see if studies with high or low df have generally larger ranges:

low_df	avg_rng	sd_rng	Q05	Q95	n	ratio
high	0.05	0.06	0.00	0.14	34	1.00
low	0.07	0.05	0.02	0.15	16	1.36

The low df studies tend to have higher ranges than the high df studies.

Variation in Estimated Standard Errors

Studies with a high ratio of SE estimates

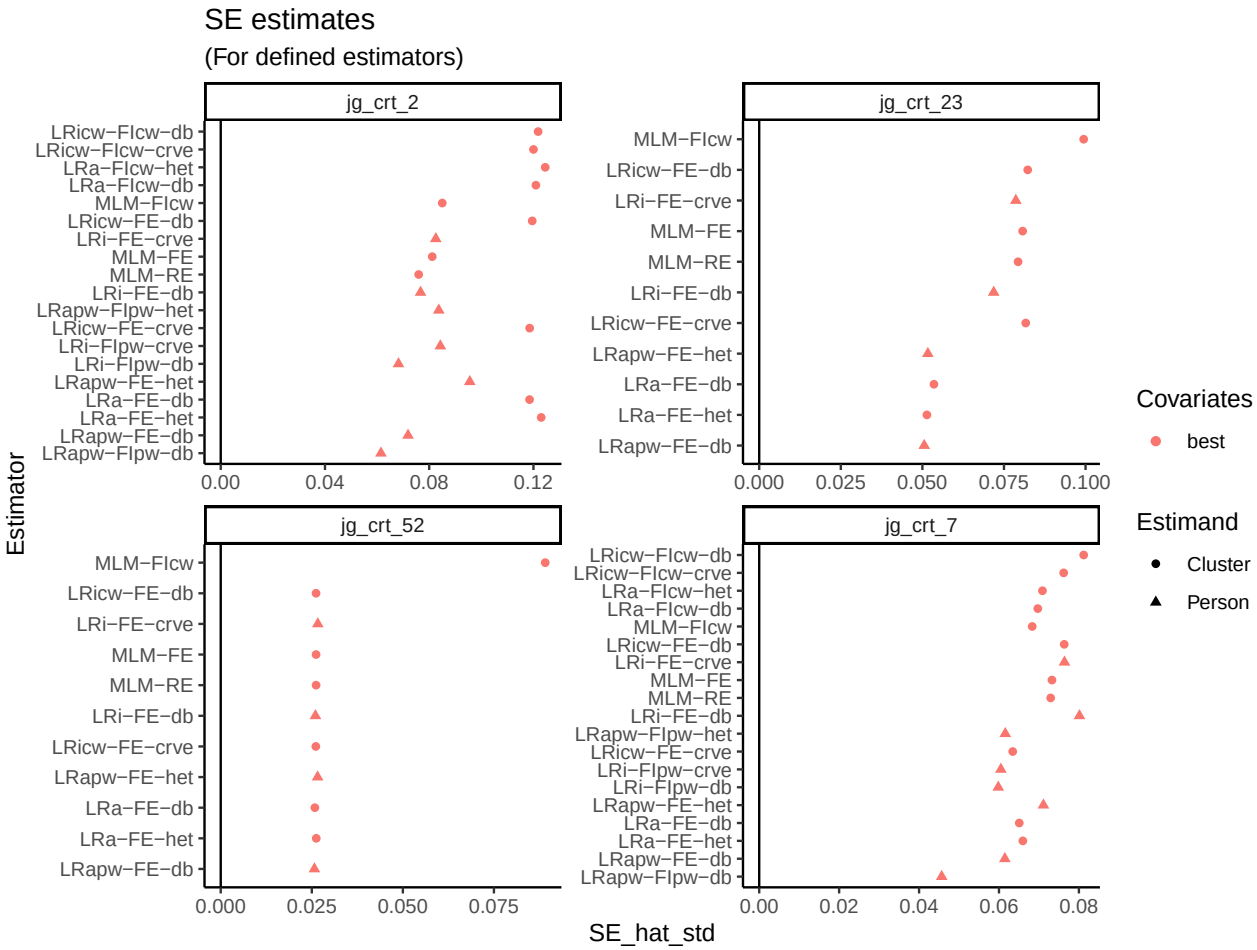
Here are all the studies with an min-max SE ratio above 1.75.

Project	UniqueID	K	J	n	cons_df	R_se	R_se_Clu	R_se_Per
Kim/set_2	jg_crt_2	7	83	2174	69	2.03	1.64	1.55
Bang/set_16	jg_crt_23	16	41	926	9	1.97	1.94	1.56
Abaluck/set_35	jg_crt_52	254	508	102869	0	3.46	3.45	1.04
Kim/set_7	jg_crt_7	7	30	1386	16	1.78	1.28	1.75

Some extra info on these studies:

UniqueID	cons_df	num_singletons	SE_min	SE_max	n_est	n_def
jg_crt_2	69	0	0.0615	0.1245	19	19
jg_crt_23	9	25	0.0506	0.0994	19	11
jg_crt_52	0	508	0.0257	0.0891	19	11
jg_crt_7	16	0	0.0457	0.0812	19	19

The individual SE estimates for the troublesome studies:



How often are the SE estimates defined?

Here we look at 'best' and see how often the SE estimates are defined for each outcome. We see that for 34 outcomes, 57% of our estimators are defined and for 16 outcomes all of them are defined.

```
##
##        0.578947368421053  1
##   best                34 16
```

How much do SE ratios go down within Estimand?

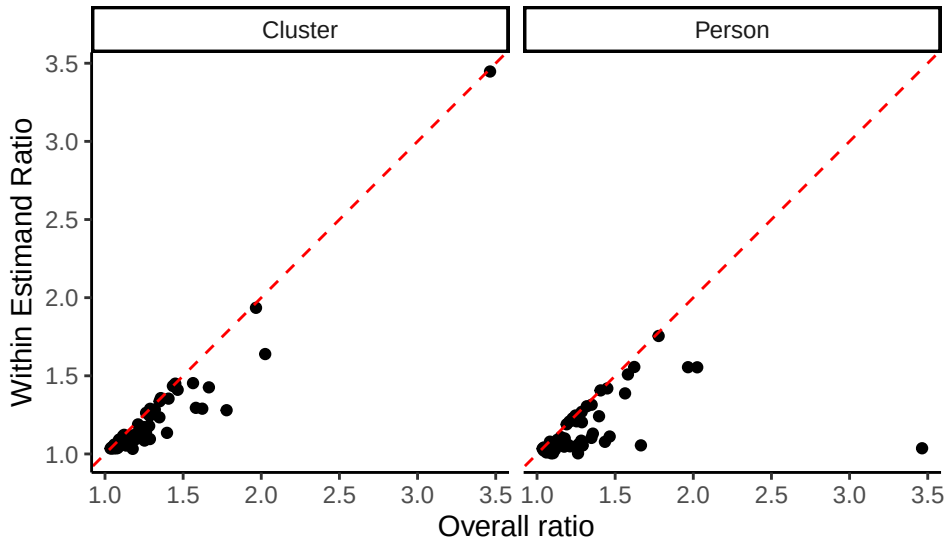
In the following we look at how much the SE ratios shrink going within estimand.

```
## # A tibble: 100 x 3
##    UniqueID                 Estimand Ratio
##    <chr>                    <chr>    <dbl>
##  1 Beacon_S8TGPA            Cluster   1.04
##  2 Beacon_S8TGPA            Person    1.03
##  3 Beacon_S8TOTCR           Cluster   1.06
##  4 Beacon_S8TOTCR           Person    1.01
##  5 Beacon_SRPRST2           Cluster   1.04
##  6 Beacon_SRPRST2           Person    1.02
##  7 CARES_DC_CST_COMPTOT_PTH Cluster   1.28
##  8 CARES_DC_CST_COMPTOT_PTH Person    1.05
##  9 CARES_DC_EMIT_PTH        Cluster   1.41
## 10 CARES_DC_EMIT_PTH        Person    1.11
## # i 90 more rows
```

UniqueID	Estimand	Ratio.x	Range	Ratio.y
Beacon_S8TGPA	Cluster	1.04	0.01	1.04
Beacon_S8TGPA	Person	1.03	0.01	1.04
Beacon_S8TOTCR	Cluster	1.06	0.01	1.06
Beacon_S8TOTCR	Person	1.01	0.01	1.06
Beacon_SRPRST2	Cluster	1.04	0.00	1.04
Beacon_SRPRST2	Person	1.02	0.00	1.04
CARES_DC_CST_COMPTOT_PTH	Cluster	1.28	0.08	1.29
CARES_DC_CST_COMPTOT_PTH	Person	1.05	0.08	1.29
CARES_DC_EMIT_PTH	Cluster	1.41	0.04	1.46
CARES_DC_EMIT_PTH	Person	1.11	0.04	1.46
CARES_DC_PT_Incred	Cluster	1.13	0.04	1.21
CARES_DC_PT_Incred	Person	1.21	0.04	1.21
CARES_DC_PT_Tools_Mind	Cluster	1.43	0.04	1.43
CARES_DC_PT_Tools_Mind	Person	1.08	0.04	1.43
CARES_TR_BPI_Incred	Cluster	1.26	0.14	1.26
CARES_TR_BPI_Incred	Person	1.06	0.14	1.26
CARES_TR_SSRS_PTH	Cluster	1.43	0.09	1.66
CARES_TR_SSRS_PTH	Person	1.06	0.09	1.66
CARES_TR_WRS_Incred	Cluster	1.14	0.11	1.26
CARES_TR_WRS_Incred	Person	1.00	0.11	1.26
CARES_TR_WRS_Tools_Mind	Cluster	1.19	0.05	1.21
CARES_TR_WRS_Tools_Mind	Person	1.05	0.05	1.21
CLC_GRCCSS_POST	Cluster	1.30	0.06	1.58
CLC_GRCCSS_POST	Person	1.51	0.06	1.58
CLC_PCCRERCORE	Cluster	1.36	0.05	1.36
CLC_PCCRERCORE	Person	1.13	0.05	1.36

UniqueID	Estimand	Ratio.x	Range	Ratio.y
EMERGE_eclsb2_BB	Cluster	1.05	0.08	1.06
EMERGE_eclsb2_BB	Person	1.03	0.08	1.06
EMERGE_eclsb2_BBSSR	Cluster	1.05	0.04	1.10
EMERGE_eclsb2_BBSSR	Person	1.02	0.04	1.10
EMERGE_team2___BB	Cluster	1.12	0.06	1.12
EMERGE_team2___BB	Person	1.03	0.06	1.12
EMERGE_team2___BBSSR	Cluster	1.10	0.04	1.20
EMERGE_team2___BBSSR	Person	1.06	0.04	1.20
MPC_eclsb2_scale	Cluster	1.12	0.04	1.12
MPC_eclsb2_scale	Person	1.07	0.04	1.12
MPC_wj_ap2	Cluster	1.12	0.05	1.17
MPC_wj_ap2	Person	1.05	0.05	1.17
MathPD_MTSS	Cluster	1.14	0.14	1.23
MathPD_MTSS	Person	1.23	0.14	1.23
MathPD_MTSS_FRACTDEC	Cluster	1.17	0.13	1.25
MathPD_MTSS_FRACTDEC	Person	1.25	0.13	1.25
MathPD_MTSS_PCTRATIO	Cluster	1.11	0.14	1.19
MathPD_MTSS_PCTRATIO	Person	1.19	0.14	1.19
PowerTeaching_TSTSSTAND	Cluster	1.10	0.09	1.29
PowerTeaching_TSTSSTAND	Person	1.20	0.09	1.29
Success_S14WJLWI_SCORE	Cluster	1.11	0.01	1.15
Success_S14WJLWI_SCORE	Person	1.11	0.01	1.15
TRIADScale-up_rema2	Cluster	1.45	0.04	1.45
TRIADScale-up_rema2	Person	1.42	0.04	1.45
TRIAD_rema2	Cluster	1.14	0.06	1.40
TRIAD_rema2	Person	1.24	0.06	1.40
TennStar_gktmathss	Cluster	1.04	0.02	1.08
TennStar_gktmathss	Person	1.08	0.02	1.08
TennStar_gktreadss	Cluster	1.07	0.02	1.13
TennStar_gktreadss	Person	1.09	0.02	1.13
jg_crt_10	Cluster	1.09	0.01	1.09
jg_crt_10	Person	1.03	0.01	1.09
jg_crt_11	Cluster	1.24	0.14	1.29
jg_crt_11	Person	1.27	0.14	1.29
jg_crt_12	Cluster	1.34	0.05	1.35
jg_crt_12	Person	1.31	0.05	1.35
jg_crt_16	Cluster	1.09	0.04	1.25
jg_crt_16	Person	1.21	0.04	1.25
jg_crt_17	Cluster	1.28	0.02	1.32
jg_crt_17	Person	1.31	0.02	1.32
jg_crt_18	Cluster	1.35	0.01	1.41
jg_crt_18	Person	1.41	0.01	1.41
jg_crt_19	Cluster	1.45	0.04	1.56
jg_crt_19	Person	1.39	0.04	1.56
jg_crt_2	Cluster	1.64	0.30	2.03
jg_crt_2	Person	1.55	0.30	2.03
jg_crt_23	Cluster	1.94	0.02	1.97
jg_crt_23	Person	1.56	0.02	1.97
jg_crt_29	Cluster	1.09	0.00	1.09
jg_crt_29	Person	1.00	0.00	1.09
jg_crt_30	Cluster	1.10	0.00	1.10
jg_crt_30	Person	1.01	0.00	1.10

UniqueID	Estimand	Ratio.x	Range	Ratio.y
jg_crt_31	Cluster	1.05	0.01	1.14
jg_crt_31	Person	1.06	0.01	1.14
jg_crt_34	Cluster	1.04	0.02	1.04
jg_crt_34	Person	1.04	0.02	1.04
jg_crt_41	Cluster	1.04	0.01	1.06
jg_crt_41	Person	1.02	0.01	1.06
jg_crt_52	Cluster	3.45	0.02	3.46
jg_crt_52	Person	1.04	0.02	3.46
jg_crt_56	Cluster	1.03	0.09	1.18
jg_crt_56	Person	1.10	0.09	1.18
jg_crt_6	Cluster	1.23	0.02	1.35
jg_crt_6	Person	1.10	0.02	1.35
jg_crt_63	Cluster	1.29	0.02	1.29
jg_crt_63	Person	1.05	0.02	1.29
jg_crt_68	Cluster	1.18	0.02	1.28
jg_crt_68	Person	1.08	0.02	1.28
jg_crt_69	Cluster	1.04	0.02	1.08
jg_crt_69	Person	1.05	0.02	1.08
jg_crt_7	Cluster	1.28	0.12	1.78
jg_crt_7	Person	1.75	0.12	1.78
jg_crt_8	Cluster	1.29	0.19	1.62
jg_crt_8	Person	1.56	0.19	1.62



This is the average reduction across outcomes:

Estimand	mean_rat	sd_rat
Cluster	0.94	0.07
Person	0.90	0.12

Are SEs for cluster weight smaller than individual?

We fit a multilevel model to our raw estimated standard errors, having a random intercept for the outcome and for the estimator. We also allow for a "random slope" for estimand within outcome, as it is possible that

some studies are better at targeting person vs cluster average effects. We finally allow a random slope for studies with singletons, to see if some estimators have estimated SEs that change more depending on whether there are singletons present:

```
## lmer(formula = log(SE_hat_std) ~ Estimand + Covariates + has_single +
##     I(log(1 + cons_df)) + I(log(nbar)) + I(log(J)) + (1 + has_single |
##     Estimator) + (1 + Estimand | UniqueID), data = combined)
##                      coef.est coef.se
## (Intercept)           0.24     0.27
## EstimandPerson       -0.03     0.02
## Covariatesnone        0.17     0.02
## Covariatesall        -0.10     0.02
## has_singlesingetons -0.22     0.10
## I(log(1 + cons_df))  0.00     0.04
## I(log(nbar))         -0.25     0.04
## I(log(J))            -0.44     0.05
##
## Error terms:
##  Groups    Name                Std.Dev. Corr
##  UniqueID  (Intercept)         0.28
##            EstimandPerson      0.02     -1.00
##  Estimator (Intercept)         0.00
##            has_singlesingetons 0.06      NaN
##  Residual                      0.41
## ---
## number of obs: 1965, groups: UniqueID, 50; Estimator, 19
## AIC = 2302.2, DIC = 2193.4
## deviance = 2232.8
```

We see no evidence that targeting person estimands is different from cluster estimands in terms of precision. There is some variation in average uncertainty by estimator.

We can look at the exponentiated coefficients to see the multiplicative effect on the SE:

	x
(Intercept)	1.27
EstimandPerson	0.97
Covariatesnone	1.18
Covariatesall	0.90
has_singlesingetons	0.80
I(log(1 + cons_df))	1.00
I(log(nbar))	0.78
I(log(J))	0.65

Relative to 'best', not controlling for anything means estimated SEs are 18% larger. Controlling for everything makes them 10% smaller. Bigger studies in terms of either n or J have smaller SEs, as expected. The estimated SEs for the studies with singletons are generally smaller; it is unclear what that is coming from.

We can dig into which estimators are better or worse:

estimator	avg	singletons
LRa-FE-db	1	1.00
LRa-FE-het	1	1.01
LRa-FIcw-db	1	1.00

estimator	avg	singletons
LRa-FIcw-het	1	1.00
LRapw-FE-db	1	0.98
LRapw-FE-het	1	1.01
LRapw-FIpw-db	1	1.00
LRapw-FIpw-het	1	1.00
LRi-FE-crve	1	0.99
LRi-FE-db	1	1.02
LRi-FIpw-crve	1	1.00
LRi-FIpw-db	1	1.00
LRicw-FE-crve	1	0.88
LRicw-FE-db	1	1.04
LRicw-FIcw-crve	1	1.00
LRicw-FIcw-db	1	1.00
MLM-FE	1	1.01
MLM-FIcw	1	1.06
MLM-RE	1	1.00

The estimators have no variation in the average estimated SE, but do differ when faced with studies with singletons. In particular, MLM-FIcw is 6% larger, and LRicw-FE-crve is 12% smaller—the latter is likely due to poorly estimated CRVEs when there are singletons.

Which estimators tend to be most extreme?

Here we tag when each estimator is either the 1st or second smallest or 1st or second largest SE estimate of all availables for that outcome (within the "best" covariate set).

Estimator	Estimand	n_def	n_extreme	per
LRa-FIcw-db	Cluster	7	0	0.00
LRi-FIpw-crve	Person	7	0	0.00
MLM-FE	Cluster	25	2	0.08
MLM-RE	Cluster	25	4	0.16
LRa-FE-db	Cluster	25	5	0.20
LRi-FE-crve	Person	25	5	0.20
LRa-FIcw-het	Cluster	7	2	0.29
LRapw-FIpw-het	Person	7	2	0.29
LRi-FIpw-db	Person	7	2	0.29
LRicw-FIcw-crve	Cluster	7	2	0.29
LRicw-FIcw-db	Cluster	7	2	0.29
LRapw-FE-het	Person	25	8	0.32
LRa-FE-het	Cluster	25	9	0.36
LRicw-FE-crve	Cluster	25	9	0.36
LRicw-FE-db	Cluster	25	9	0.36
LRi-FE-db	Person	25	10	0.40
MLM-FIcw	Cluster	25	11	0.44
LRapw-FE-db	Person	25	14	0.56
LRapw-FIpw-db	Person	7	4	0.57

The `n_def` column is the number of outcomes where the estimator was even defined.

I'm not sure this table is particularly informative, but it does show that some estimators are more likely to be extreme than others. The problem is some estimators aren't even defined, which robs them of the

opportunity to be informative.